SpaceX Launch Analysis

Launch Success Rate and Launch Frequency Over Time

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1 Introduction

Since its first launches in 2006, SpaceX has developed its launch capabilities and increased the number of missions it performs. This report checks whether these developments were accompanied by improvements in launch reliability.

The main research question is: *How has SpaceX's orbital launch success rate changed over time?*

The secondary research question is: *How has the number of orbital launches per year changed over the same period?*

The analysis focuses on the period 2006–2022.

2 Dataset and Preparation

The analysis uses the file `spacex_launches_20251222.csv`. The SpaceX Mission Archive (2006–2022), is available on Kaggle, as the original data source.

Two variables were used: the UTC launch date and the launch outcome. The dates were converted into years, and launches with unknown outcomes were excluded. The remaining launches were classified as successful or unsuccessful and grouped by year.

3 Methodology

For each year, the total number of launches and the number of successful launches were calculated. These values were used to determine the annual success rate:

$$\text{Success rate} = \frac{\text{successful launches}}{\text{total launches}} \times 100\%.$$

The mean, median and standard deviation of the annual success rates were also calculated. Two graphs were created to examine changes in success rate and launch frequency.

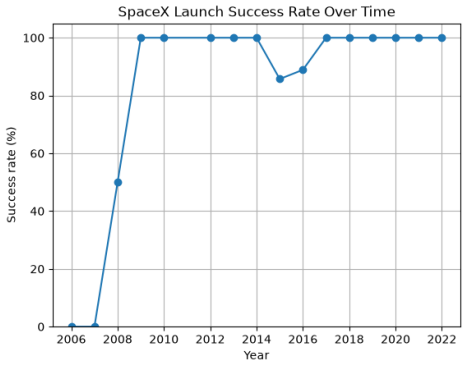


Figure 1: Annual SpaceX launch success rate between 2006 and 2022.

4 Launch Success Rate

Figure 1 shows that launch success rates varied considerably in the early years. Because relatively few launches were performed, individual failures had a large effect on the annual percentages.

Success rates generally improved after 2008, although a decrease occurred in 2015. On 28 June 2015, a Falcon 9 rocket failed approximately two minutes after liftoff during the CRS-7 mission to the International Space Station. Launches were temporarily suspended while the failure was investigated [2].

In later years, the success rate became more stable and remained consistently high.

5 Launch Frequency

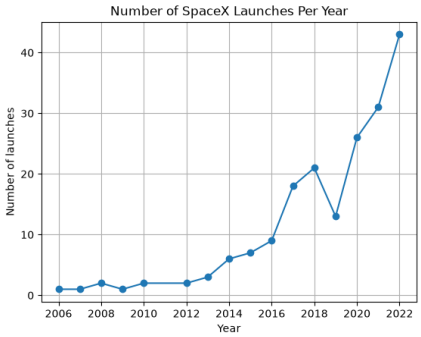


Figure 2: Annual number of SpaceX launches between 2006 and 2022.

Figure 2 shows an increase in launch activity. SpaceX initially performed only a few launches per year, but the number increased more rapidly after 2014.

An exception occurred in 2019, when the number of launches decreased compared with 2018. The presentation discusses possible contributing factors, including changes in satellite demand, development of Starship and Starlink, work on Crew Dragon for NASA, and workforce reductions. These possible explanations cannot be confirmed from the graph alone [3].

Launch activity increased again after 2019 and reached its highest recorded level in the dataset in 2022.

6 Statistical Results

The calculated descriptive statistics for the annual launch success rates were:

Statistic	**Result**
Mean success rate	82.8%
Median success rate	100%
Standard deviation	34.7 percentage points

The mean indicates an average annual success rate of 82.8%. The median of 100% shows that at least half of the years had a perfect success rate. The relatively large standard deviation reflects the variation in success rates, especially during the early years.

These statistics give equal weight to each year rather than to each individual launch.

7 Conclusion

The results indicate that SpaceX improved its orbital launch reliability while also increasing its launch frequency between 2006 and 2022.

Although success rates were less stable during the early years, they became consistently high in the later period. At the same time, the number of launches increased substantially, despite a temporary decrease in 2019.

Overall, the analysis suggests that SpaceX was able to expand its launch activity while maintaining high success rates.

8 References

References

[1] SpaceX Mission Archive (2006–2022). Kaggle dataset.
[2] Evans, B. (2015). *2015 in Review: A Year of Success, Disappointment and Recovery.* AmericaSpace.
[3] Wall, M. (2022). *SpaceX Hasn't Flown a Falcon Heavy Rocket Since 2019. Here's Why.* Space.com.